

HANDS-ON ELECTRONIC STRUCTURE

Quantum ESPRESSO

A practical first tutorial

SCF calculations, convergence, bands, magnetism
and structural optimization

Adapted from the supplied `handson_pwscf.pdf`
with current input syntax and worked explanations

The route through this tutorial

Material	What you will learn
Silicon	Build an input, reach self-consistency, test numerical convergence, fit a lattice parameter, calculate bands
Aluminum	Treat partially occupied bands and test smearing with the k -point grid
Iron	Set spin polarization, compare magnetic states, calculate DOS/PDOS and optimize an hcp cell
Carbon	Model a 2D sheet, construct a supercell, relax graphane and an epoxide defect

Work through one material at a time. Record the input settings and the quantity you are trying to converge.

Before the first calculation

You need a Quantum ESPRESSO installation with `pw.x`, `bands.x`, `dos.x` and `projwfc.x`, plus suitable UPF pseudopotentials.

```
mkdir -p qe_tutorial/pseudo qe_tutorial/tmp
cd qe_tutorial
which pw.x
```

- Run all commands here so `pseudo_dir='./pseudo'` and `outdir='./tmp'` resolve consistently.
- Place validated UPF files in `pseudo/`. The short names `Si.UPF`, `Al.UPF`, `Fe.UPF`, `C.UPF`, `H.UPF`, `O.UPF` are **local aliases**, not download names.
- Record each original filename, source, XC functional, valence count and recommended cutoffs. Use compatible XC choices.

Reference: [QE pseudopotentials and linked libraries](#)

What is supplied, and what you must choose

- This deck is self-contained as a $\text{\LaTeX}$ document. Its examples do not depend on the old tutorial archive.
- The original `handson_pwscf.tgz`, calculation outputs and UPF files were not attached. Historical energies are not new benchmark results.
- The numerical cutoffs, grids, smearing widths and cell sizes below are **starting examples**. Convergence depends on the dataset, material and target observable.
- Examples use ordinary semilocal DFT with scalar-relativistic pseudopotentials, without spin-orbit coupling. Fe examples assume PBE.

The syntax follows the online QE 7.5 documentation. If your installed version differs, consult its own `Doc/INPUT_*.html` files.

The model behind pw.x

Density-functional theory (DFT) describes the electronic ground state using the electron density $n(\mathbf{r})$.

$$\hat{H}_{\text{KS}}[n] \psi_{n\mathbf{k}} = \epsilon_{n\mathbf{k}} \psi_{n\mathbf{k}}.$$

- Kohn–Sham orbitals $\psi_{n\mathbf{k}}$ are auxiliary one-electron states. Their occupations build the density.
- A **pseudopotential** replaces the nucleus and core-electron effects with an effective ionic potential. Valence electrons remain explicit.
- **Plane waves** form the basis for the orbitals. The cell repeats periodically in all three directions.
- LDA and PBE are approximate exchange–correlation (XC) functionals. Numerical convergence does not remove their model error.

Why a self-consistent field loop is needed

The Hamiltonian depends on the density, but the density depends on its orbitals.

$$n^{(i)} \longrightarrow \hat{H}[n^{(i)}] \longrightarrow \{\psi, \epsilon, f\} \longrightarrow n_{\text{out}}^{(i)}.$$

A simple picture of density mixing is

$$n^{(i+1)} = (1 - \beta)n^{(i)} + \beta n_{\text{out}}^{(i)}.$$

- Repeat until the input and output densities agree to the requested accuracy.
- `conv_thr` sets a threshold on QE's estimated SCF energy error, in Ry.
- `mixing_beta` affects iteration stability. A smaller value may help oscillating calculations, but can slow convergence.

SCF keeps the nuclei fixed. Geometry optimization adds an outer loop that moves nuclei after converged electronic steps.

Units and coordinates

Input	Meaning
<code>ecutwfc</code> , <code>ecutrho</code> , <code>degauss</code>	Energies in Ry
<code>celldm(1)</code>	Lattice scale in bohr
<code>A</code> , <code>B</code> , <code>C</code>	Lattice lengths in Å
<code>ATOMIC_POSITIONS</code> <code>angstrom</code>	Cartesian positions in Å
<code>ATOMIC_POSITIONS</code> <code>alat</code>	Cartesian positions divided by the lattice scale
<code>ATOMIC_POSITIONS</code> <code>crystal</code>	Fractions of the primitive lattice vectors

$$1 \text{ bohr} \simeq 0.529177 \text{ Å}, \quad 1 \text{ Ry} \simeq 13.605693 \text{ eV}.$$

Use either `celldm(...)` or `A,B,C,...`, never both. Always write the coordinate-unit label explicitly.

A cell in real space and a grid in reciprocal space

The lattice vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ define the repeating cell. Fractional coordinates (s_1, s_2, s_3) mean

$$\mathbf{r} = s_1 \mathbf{a}_1 + s_2 \mathbf{a}_2 + s_3 \mathbf{a}_3.$$

Reciprocal vectors satisfy

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}, \quad \mathbf{k} = \kappa_1 \mathbf{b}_1 + \kappa_2 \mathbf{b}_2 + \kappa_3 \mathbf{b}_3.$$

- A periodic solid has states across a Brillouin zone (BZ). A weighted k -point sum approximates a BZ integral.
- Symmetry can reduce the full mesh to an **irreducible** set. Weights account for the equivalent points.
- A larger real-space cell has a smaller reciprocal-space cell.

The anatomy of a pw.x input

```
&CONTROL
  calculation = 'scf'
  ...
/
&SYSTEM
  ...
/
&ELECTRONS
  ...
/
ATOMIC_SPECIES
...
ATOMIC_POSITIONS crystal
...
K_POINTS automatic
...
```

Namelists contain settings:

- **&CONTROL**: task and file locations
- **&SYSTEM**: cell, atoms and physical model
- **&ELECTRONS**: SCF settings

Cards contain lists, such as species, atomic coordinates and *k*-points.

A slash ends each namelist. Use plain ASCII quotes.

The dots here only indicate omitted content.

Silicon: a two-atom primitive cell

Diamond silicon has an fcc Bravais lattice with a two-atom basis. The conventional cubic cell contains eight atoms, but the primitive cell contains two.

$$V_{\text{primitive}} = \frac{a^3}{4}, \quad \mathbf{d}_1 = (0, 0, 0), \quad \mathbf{d}_2 = (a/4, a/4, a/4).$$

- `ibrav=2` selects QE's fcc primitive vectors.
- `nat=2` counts atoms in the simulated cell.
- `ntyp=1` counts distinct species labels.
- With four valence electrons per Si, the cell has eight electrons and four doubly occupied bands in the nonmagnetic ground state.

A species label links an atom to its mass and pseudopotential. The valence-electron count comes from the UPF file, not the atomic number.

Silicon input: calculation settings

Create `si.scf.in`. This block and the next slide form one complete input.

```
&CONTROL
  calculation = 'scf', prefix = 'si'
  pseudo_dir = './pseudo', outdir = './tmp'
  tprnfor = .true., tstress = .true.
/
&SYSTEM
 ibrav = 2, A = 5.43, nat = 2, ntyp = 1
ecutwfc = 60.0, ecutrho = 240.0
occupations = 'fixed'
/
&ELECTRONS
  conv_thr = 1.0d-8
  mixing_beta = 0.5
/
```

Assumption: a four-valence-electron, norm-conserving Si UPF. The 60/240 Ry cutoffs are starting values. For a different dataset, use its recommended values and test them.

Silicon input: atoms and sampling

Append these cards to `si.scf.in`:

```
ATOMIC_SPECIES
Si 28.085 Si.UPF
ATOMIC_POSITIONS alat
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25
K_POINTS automatic
6 6 6 1 1 1
```

- `alat` means Cartesian coordinates in units of $a = 5.43 \text{ \AA}$. The second atom is at $(1.3575, 1.3575, 1.3575) \text{ \AA}$.
- The six mesh numbers mean $6 \times 6 \times 6$ divisions and a half-grid shift in each reciprocal direction.
- These atomic coordinates are **not** fractions of QE's fcc primitive vectors.

Running silicon and finding the result

```
pw.x -in si.scf.in > si.scf.out 2>&1  
grep -E 'convergence|JOB DONE' si.scf.out  
tail -n 20 si.scf.out
```

- Read the header: QE version, pseudopotential, XC functional, number of electrons, bands, k -points and cutoffs.
- Check both **electronic convergence** and normal termination. JOB DONE. alone does not prove a relaxation converged.
- The line beginning with ! reports the final SCF total energy in Ry per simulated cell.
- Keep tmp/si.save and required wavefunctions for later steps. Never run concurrent jobs with the same prefix and outdir.

For MPI, use the launcher prescribed by your machine, for example `mpirun -np 4 pw.x -in si.scf.in`. Exact scratch filenames depend on version and options.

SCF convergence and numerical convergence

Question	How to answer it
SCF convergence	At fixed input settings, has the density iteration reached its threshold?
Basis convergence	Does increasing <code>ecutwfc</code> , and where needed <code>ecutrho</code> , change the target quantity negligibly?
BZ convergence	Does a denser k -mesh change the target quantity negligibly?
Model accuracy	Do the XC functional, pseudopotential and physical assumptions describe the property adequately?

A tightly converged SCF calculation can still have a poor basis or a coarse k -mesh. Choose a tolerance for the quantity you will report.

Plane-wave cutoffs

Each orbital uses plane waves satisfying

$$\frac{\hbar^2 |\mathbf{k} + \mathbf{G}|^2}{2m_e} \leq E_{\text{cut}}.$$

- Higher `ecutwfc` includes shorter-wavelength detail and usually costs more memory and time.
- The density contains products of orbitals, so a norm-conserving calculation normally uses `ecutrho=4*ecutwfc`.
- Ultrasoft augmentation charges often need a higher density cutoff. PAW datasets also need their own tests.
- The ideal variational trend is downward with a larger orbital basis. Finite grids, changing density cutoffs and incomplete SCF can spoil strict monotonicity.

Converge energies, forces or stress according to the task. A cutoff adequate for a rough energy may be inadequate for a precise cell optimization.

A worked convergence decision

Illustrative arithmetic only: these values are invented to show the test, not calculated Si results.

For a two-atom cell, suppose:

ecutwfc (Ry)	E (Ry/cell)	Error vs 80 Ry (meV/atom)
40	-15.80000	2.721
60	-15.80030	0.680
80	-15.80040	0

$$\Delta E_{60} = \frac{0.00010 \times 13.605693 \times 1000}{2} = 0.6803 \text{ meV/atom.}$$

A target of 1 meV/atom accepts 60 Ry against this reference. **One further, higher-cutoff test** should confirm that the reference is stable.

A k -mesh convergence test

At a sufficiently high cutoff, try $N = 2, 4, 6, 8, 10$:

```
K_POINTS automatic
N N N 1 1 1
```

Replace each N with the same integer before running.

- Hold geometry, XC, pseudopotentials, cutoffs and SCF threshold fixed.
- Record the full mesh, offsets, irreducible count and energy per atom.
- Compare successive denser meshes and the densest available reference. k -mesh convergence need not be monotonic.
- A zero offset includes the reciprocal origin. An offset of 1 shifts that direction by half a mesh step.

The old two-special-point fcc example is reproduced by a shifted 2^3 mesh with the relevant symmetry. It is a teaching mesh, not a universal accuracy choice.

A small calculation log prevents large mistakes

For every test, record at least:

Input record	Output record
Structure and atom count	Converged or failed
UPF identifiers and XC	Final energy per cell and per atom
Wavefunction/density cutoffs	Maximum force, if relevant
Full k -mesh and offsets	Stress or pressure, if relevant
Occupation method and width	Runtime and memory, if useful
SCF threshold and QE version	Target-property difference

Use distinct prefixes for independent tests, such as `si_ecut60`. Use matching prefixes only when a later step intentionally reads the preceding calculation.

Silicon's equilibrium lattice parameter

For ideal diamond Si, symmetry makes the atomic forces vanish at every cubic a .

Zero forces therefore do not determine the equilibrium cell.

$$a_0 = \arg \min_a E(a), \quad V(a) = a^3/4.$$

1. Select converged cutoffs and a k -mesh.
2. Run a range of lattice parameters that brackets the minimum, for example 5.2 to 5.6 Å.
3. Refine the spacing near the minimum. Keep numerical settings consistent.

Room-temperature Si is approximately 5.43 Å $\simeq$ 10.26 bohr. The original pair 5.47 Å and 10.26 bohr was inconsistent.

Reference: [NIST silicon lattice-parameter standard](#)

An equation-of-state fit

A smooth fit to $E(V)$ estimates the minimum and the curvature:

$$P(V) = -\frac{dE}{dV}, \quad B_0 = V_0 \left. \frac{d^2E}{dV^2} \right|_{V_0}.$$

- V_0 : equilibrium volume. B_0 : bulk modulus, a measure of resistance to compression.
- Use points on both sides of the minimum. An excellent-looking fit cannot repair unconverged input energies.
- QE's `ev.x` supports equation-of-state fits. Follow the executable's prompts for lattice type, length/volume units and energy units.
- For this primitive fcc cell, $a_0 = (4V_0)^{1/3}$. Mixing primitive-cell and conventional-cell volumes gives the wrong result.

Compare theory and experiment with temperature and XC limitations in mind. Do not interpret agreement in one material as general validation.

SCF, bands and NSCF calculations

Task	Density/potential	Sampling and purpose
scf	Iteratively updated	BZ integration for the ground state
bands	Reads a fixed SCF potential	A path through reciprocal space for dispersion
nscf	Reads a fixed SCF potential	Usually a dense uniform mesh for DOS or other properties

- Preserve structure, pseudopotentials, model, prefix and outdir along a dependent workflow.
- Set enough nbnd to include the empty states you want to inspect.
- A path samples only selected directions. It cannot replace a uniform mesh for a DOS.

Silicon bands along an explicit path

Copy the converged Si SCF input to `si.bands.in`. Set `calculation='bands'`, add `nbnd=8`, and replace the *k*-card with:

```
K_POINTS tpiba_b
6
0.500 0.500 0.500 30
0.000 0.000 0.000 30
0.000 0.000 1.000 30
0.000 0.500 1.000 30
0.000 0.750 0.750 30
0.000 0.000 0.000 1
```

$$L - \Gamma - X - W - K - \Gamma.$$

`tpiba_b` uses Cartesian coordinates in units of $2\pi/a$. Each final integer controls interpolation toward the next vertex. These coordinates use the fcc conventional Cartesian axes of this Si example.

Separate disconnected path segments in the plot. Continuity is useful within a segment, but a band path need not be globally connected.

Band postprocessing with bands.x

Run `pw.x -in si.bands.in > si.bands.out 2>&1`. Then create `si.bands.pp.in`:

```
&BANDS
  prefix = 'si', outdir = './tmp'
  filband = 'si.bands.dat'
  lsym = .true.
/
```

```
bands.x -in si.bands.pp.in > si.bands.pp.out 2>&1
```

- `si.bands.dat.gnu` is directly plottable. Its energies are in eV.
- `lsym=.true.` enables symmetry analysis and writes `si.bands.dat.rap`.
- `lsym=.false.` disables that analysis. Band ordering can use wavefunction overlaps depending on `no_overlap`.

Reference: [QE bands.x input reference](#)

Interpreting a band plot

For a semiconductor, choose the valence-band maximum (VBM) as the energy zero. For a metal, the Fermi energy is a common reference.

```
# In gnuplot, assign Eref from your own output:  
Eref = 0.0    # replace this value  
set ylabel "Energy - Eref (eV)"  
plot "si.bands.dat.gnu" using 1:($2-Eref) with lines
```

- The horizontal coordinate follows distance along the path, not a single Cartesian k component. Add labels at the path vertices.
- A band gap is the difference between the lowest conduction and highest valence energies. The extrema may be at different k -points.
- A selected path may miss an extremum. Dense sampling is needed for an accurate gap.
- Semilocal Kohn–Sham gaps often underestimate measured excitation gaps.

Why metals need a different occupation treatment

In a metal, some bands cross the Fermi energy E_F . The occupied region changes sharply across the Brillouin zone.

$$N_e = \sum_{nk} w_k f_{nk}.$$

Here f_{nk} includes the appropriate spin occupancy.

- A coarse mesh may poorly sample the Fermi surface, making energies and forces sensitive to the grid.
- **Smearing** smooths the occupation transition over an energy width.
- Include enough bands above E_F , so the highest calculated bands have negligible occupations.
- Converge the mesh and smearing width together.

Aluminum: changes to the Si template

Save a separate `al.scf.in`. Keep `&ELECTRONS`; set `prefix='al'`. Replace the entire `&SYSTEM` block and all three cards:

```
&SYSTEM
 ibrav=2, A=4.05
nat=1, ntyp=1
ecutwfc=60, ecutrho=240
occupations='smearing'
smearing='mv'
degauss=0.02, nbnd=8
/
```

```
ATOMIC_SPECIES
Al 26.9815 Al.UPF
ATOMIC_POSITIONS crystal
Al 0.0 0.0 0.0
K_POINTS automatic
12 12 12 1 1 1
```

Assume a three-valence-electron, norm-conserving Al UPF. Adjust cutoffs for your chosen dataset.

$a = 4.05 \text{ \AA}$, the mesh and the width are initial test settings. Run `pw.x -in al.scf.in > al.scf.out 2>&1`.

What the smearing width means

QE option	Interpretation
gaussian	Simple Gaussian broadening
mv	Marzari–Vanderbilt cold smearing
mp	Methfessel–Paxton smearing
fd	Fermi–Dirac occupations

- degauss is in Ry. $0.02 \text{ Ry} \simeq 0.272 \text{ eV}$.
- Only Fermi–Dirac broadening has the direct interpretation $\text{degauss} = k_B T$. Other choices are numerical integration devices.
- A wider width can stabilize coarse meshes but changes the finite-width energy and occupations.
- Reducing the width usually requires a denser mesh. Compare the same energy convention throughout the test.

A joint mesh-and-smearing test

For Al, an illustrative test matrix is:

Full mesh	degauss (Ry)		
	0.01	0.02	0.04
8^3	run	run	run
12^3	run	run	run
16^3	run	run	run

1. At each width, check convergence with the mesh.
2. On sufficiently dense meshes, reduce the width and compare the target property.
3. Extend the matrix until changes fall below your tolerance.

Cold smearing is a useful starting method, not a guarantee. Do not assume that a particular 12^3 mesh or a width from the old tutorial is converged for your calculation.

Iron: magnetism and pseudopotentials

- Iron's localized $3d$ states can make a plane-wave calculation expensive. This example uses an **ultrasoft** Fe dataset.
- Fe does not require ultrasoft pseudopotentials. Validated norm-conserving or PAW datasets are also possible.
- `nspin=2` enables **collinear spin polarization**: separate spin-up and spin-down densities.
- `starting_magnetization` supplies an initial spin asymmetry. It does not fix the converged moment.

$$n = n_{\uparrow} + n_{\downarrow}, \quad m = n_{\uparrow} - n_{\downarrow}.$$

Spin treatment and XC are separate choices. PBE with `nspin=2` is spin-polarized GGA, not an LDA calculation.

Ferromagnetic bcc Fe: a starting input

Make `fe.scf.in` from the Si template, set `prefix='fe'`, and replace `&SYSTEM` and all cards:

```
&SYSTEM
ibrav=3, A=2.87
nat=1, ntyp=1
ecutwfc=60, ecutrho=600
nspin=2
starting_magnetization(1)=0.5
occupations='smearing'
smearing='mv', degauss=0.02
nbnd=20
/
```

```
ATOMIC_SPECIES
Fe 55.845 Fe.UPF
ATOMIC_POSITIONS crystal
Fe 0.0 0.0 0.0
K_POINTS automatic
12 12 12 1 1 1
```

Use a PBE ultrasoft Fe UPF. Check its valence count, cutoff recommendations and unoccupied-band occupations.

For `nspin=2`, `nbnd` is the number of bands in each spin channel. Twenty is an example, not a convergence result.

Total and absolute magnetization

For collinear spin densities, QE reports cell-integrated quantities:

$$M_{\text{tot}} = \mu_B \int_{\text{cell}} [n_{\uparrow}(\mathbf{r}) - n_{\downarrow}(\mathbf{r})] d^3r,$$

$$M_{\text{abs}} = \mu_B \int_{\text{cell}} |n_{\uparrow}(\mathbf{r}) - n_{\downarrow}(\mathbf{r})| d^3r.$$

- M_{tot} permits cancellation of opposite spin polarization.
- $M_{\text{abs}} \geq |M_{\text{tot}}|$. An AFM state can have zero total magnetization and a nonzero absolute magnetization.
- These are not uniquely defined atom-resolved moments.
- QE may list the same spatial k -mesh for both spin channels. This does not double its spatial sampling density.

A two-sublattice antiferromagnetic starting state

Use a conventional cubic bcc cell with `ibrav=1`, `nat=2`, `ntyp=2`, `prefix='fe_afm'` and enough bands, for example `nbnd=40`.

```
! Inside &SYSTEM:  
nspin=2  
starting_magnetization(1)= 0.5  
starting_magnetization(2)=-0.5
```

```
ATOMIC_SPECIES  
Fe1 55.845 Fe.UPF  
Fe2 55.845 Fe.UPF  
ATOMIC_POSITIONS crystal  
Fe1 0.0 0.0 0.0  
Fe2 0.5 0.5 0.5
```

- Both labels use the same element and UPF. Separate labels let the two sites start with opposite spin polarization.
- The lattice constant remains the conventional bcc a . Retest reciprocal sampling for this different cell.
- The final state may change or lose its initial order. Inspect it and compare converged **energies per atom**.

Converging an ultrasoft calculation

The ratio $D = E_\rho/E_{\text{wfc}}$ is often called the **dual**. A value around 8–12 is a common USPP starting range.

1. Choose a density cutoff high enough to represent the augmentation charge.
2. Increase `ecutwfc` at fixed high `ecutrho`.
3. At the selected orbital cutoff, increase `ecutrho`.
4. Confirm the final pair together with the k -mesh and smearing.
 - A low density cutoff can damage forces, stress and energy differences.
 - Varying the dual can be useful, but changing both cutoffs at once obscures which error matters.
 - A nonmonotonic energy curve is a warning to investigate, not evidence for a new physical effect.

Density of states: how many states occur near an energy?

The density of states is a Brillouin-zone sum:

$$D(E) = \sum_{nk\sigma} w_k \delta(E - \epsilon_{nk\sigma}).$$

- The delta function places a contribution at each eigenvalue. Integration or broadening turns the discrete sample into a curve.
- A band path has biased sampling and unsuitable weights for this task.
- Workflow: converged Fe SCF, dense uniform-grid NSCF, then `dos.x`.
- In `fe.nscf.in`, retain the Fe model and paths, set `calculation='nscf'`, increase `nbnd` if needed, and test a denser mesh such as 20^3 .

Keep the saved NSCF data and wavefunctions until both DOS and PDOS are complete. Do not overwrite them with an unrelated bands calculation.

Fe DOS with explicit Gaussian broadening

Create `fe.dos.in`. This route makes a direct comparison with PDOS simple.

```
&DOS
  prefix='fe', outdir='./tmp', fildos='fe.dos.dat'
  Emin=-20.0, Emax=30.0, DeltaE=0.02
  ngauss=0, degauss=0.01
/
```

```
pw.x -in fe.nscf.in > fe.nscf.out 2>&1
dos.x -in fe.dos.in > fe.dos.out 2>&1
```

- `Emin`, `Emax` and `DeltaE` are in **eV**. Adjust the example window to cover your calculated bands.
- `degauss` is in **Ry**. Test width and mesh until the desired features are stable.
- For collinear Fe, the DOS file has energy, spin-up DOS, spin-down DOS and integrated DOS. Read its header and shift energy by E_F when plotting.

Reference: [QE dos.x input reference](#)

Tetrahedra are an alternative integration route

For a bulk DOS:

1. Use a dense automatic, symmetry-compatible mesh in the NSCF input.
 2. Set `occupations='tetrahedra'` and remove the NSCF smearing settings.
 3. Omit `degauss` from the `dos.x` input to use the tetrahedron information.
- QE's tetrahedra option is the Blöchl scheme. `tetrahedra_lin` selects the original linear variant.
 - Current `projwfc.x` also supports tetrahedron PDOS. The old claim that tetrahedra are unimplemented is obsolete.
 - Different postprocessors can use different tetrahedron variants. For a simple DOS–PDOS comparison, use matching Gaussian settings.

Neither method removes the need for mesh convergence. A finer postprocessing energy grid does not generate additional electronic eigenvalues.

Projected DOS: which orbitals contribute?

`projwfc.x` projects Kohn–Sham states onto atomic-like orbitals supplied by the pseudopotentials. Create `fe.pdos.in`:

```
&PROJWFC
  prefix='fe', outdir='./tmp', filpdos='fe.pdos'
  Emin=-20.0, Emax=30.0, DeltaE=0.02
  ngauss=0, degauss=0.01
/
```

```
projwfc.x -in fe.pdos.in > fe.pdos.out 2>&1
```

- Use the same NSCF dataset, energy grid and Gaussian broadening as the DOS comparison.
- Inspect `fe.pdos.out` to identify atomic and orbital indices. A filename like `wfc#2(d)` depends on the UPF.
- Atomic projectors need not span every plane-wave state. Their summed PDOS can differ from the total DOS.

Reference: [QE projwfc.x input and output reference](#)

Reading spin-resolved DOS and PDOS files

For a collinear magnetic calculation, `fe.pdos.pdos_tot` contains:

Column	Meaning
1	Energy in eV
2, 3	Total DOS, spin up and spin down
4, 5	Sum of projected DOS, spin up and spin down

- Individual orbital files contain spin-resolved sums followed by resolved components. Always check the header.
- Both spin channels have nonnegative DOS in a Gaussian plot. Drawing spin-down below zero is only a display convention.
- A large projection mismatch may indicate missing orbital character. Compare identical sampling and integration settings before interpreting it.

hcp iron: two lattice parameters

For QE's hexagonal cell,

$$\mathbf{a}_1 = (a, 0, 0), \quad \mathbf{a}_2 = (-a/2, \sqrt{3}a/2, 0), \quad \mathbf{a}_3 = (0, 0, c).$$

```
! Inside &SYSTEM:  
ibrav=4, A=2.50, C=4.10  
nat=2, ntyp=1  
  
ATOMIC_POSITIONS crystal  
Fe 0.0000000 0.0000000 0.0  
Fe 0.3333333 0.6666667 0.5
```

$$V = \frac{\sqrt{3}}{2} a^2 c.$$

The second atom is at

$$(0, a/\sqrt{3}, c/2).$$

Starting lengths only. Optimize the cell and check the magnetic state.

Equivalent input: `celldm(1)=a/0.529177` and `celldm(3)=c/a`, with a in Å. Or use `ibrav=0` and explicit `CELL_PARAMETERS` angstrom.

An hcp equation of state at fixed volume

At each selected V , minimize the energy with respect to the cell shape. For a ratio $r = c/a$,

$$a = \left(\frac{2V}{\sqrt{3}r} \right)^{1/3}, \quad c = ra.$$

You can scan r , or use `calculation='vc-relax'` with:

```
&IONS
  ion_dynamics='bfgs'
/
&CELL
  cell_dynamics='bfgs'
  cell_dofree='shape'
/
```

Append these namelists after `&ELECTRONS`. `shape` fixes volume but permits shape changes. Preserve/check hexagonal symmetry if you want only the hcp branch. Isotropic residual stress may remain at a nonequilibrium volume.

Comparing crystal phases of iron

- Converge and optimize each structure and each relevant magnetic state using consistent pseudopotentials and XC.
- Compare energy **per atom**, since bcc, fcc and hcp input cells may contain different numbers of atoms.
- At a specified pressure, the static-lattice comparison uses enthalpy:

$$H = E + PV.$$

- A single relaxed calculation does not establish the most stable magnetic branch or the finite-temperature phase diagram.

Cell optimization also requires converged stress. Basis changes as the cell changes can introduce Pulay stress, so verify the final geometry with tighter cutoffs.

A 2D sheet in a 3D periodic calculation

Graphene is a honeycomb lattice: a hexagonal Bravais lattice with two carbon atoms.

$$a \simeq 2.46 \text{ \AA}, \quad d_{\text{C-C}} = a/\sqrt{3} \simeq 1.42 \text{ \AA}.$$

- Put the sheet in the xy plane and add vacuum along z . The calculation still repeats in all three directions.
- Start with a cell height such as $C = 20 \text{ \AA}$, then test larger heights. This is not a universal converged vacuum.
- Use a dense in-plane mesh and one k_z point when the isolated-layer limit is adequate.
- Keep the vacuum height fixed during in-plane relaxation. Unrestricted 3D cell relaxation can collapse the vacuum.

The original $C = 7.389 \text{ \AA}$ is a compact teaching example. Verify layer separation for the property and any electrostatic corrections you use.

Graphene input: calculation settings

Create `graphene.scf.in`. Combine this with the next slide.

```
&CONTROL
  calculation='scf', prefix='graphene'
  pseudo_dir='./pseudo', outdir='./tmp'
  tprnfor=.true., tstress=.true.
/
&SYSTEM
 ibrav=4, A=2.46, C=20.0, nat=2, ntyp=1
  ecutwfc=60, ecutrho=240, nbnd=8
  occupations='smearing', smearing='mv', degauss=0.005
/
&ELECTRONS
  conv_thr=1.0d-8, mixing_beta=0.4
/
```

Assume a four-valence-electron, norm-conserving carbon UPF. Test cutoff, mesh, vacuum and width. A tiny gap from sparse sampling can be a numerical artifact.

Graphene input: positions and k -mesh

Append the cards:

```
ATOMIC_SPECIES
C 12.011 C.UPF
ATOMIC_POSITIONS crystal
C 0.0000000000 0.0000000000 0.5
C 0.3333333333 0.6666666667 0.5
K_POINTS automatic
18 18 1 0 0 0
```

- The sheet lies at $z = C/2 = 10 \text{ \AA}$, centered in the cell.
- With the hexagonal vectors on the hcp slide, $(1/3, 2/3, 0)$ gives Cartesian $(0, a/\sqrt{3}, 0)$.
- The unshifted mesh contains Γ . In this convention, meshes divisible by three also sample the K point.

Run `pw.x -in graphene.scf.in > graphene.scf.out 2>&1`. Inspect electron count, SCF convergence and the geometry printed by QE.

Graphene bands through the Dirac point

After a converged SCF, copy the input to `graphene.bands.in`, change to `calculation='bands'`, and use:

```
K_POINTS crystal_b
4
0.0000000000 0.0000000000 0.0 40
0.5000000000 0.0000000000 0.0 40
0.3333333333 0.3333333333 0.0 40
0.0000000000 0.0000000000 0.0 1
```

$$\Gamma - M - K - \Gamma.$$

- `crystal_b` uses fractions of this cell's reciprocal vectors.
- Reuse `bands.x` with `prefix='graphene'`. Near K , the branches meet in the ideal model.
- A 1D path gives a cut through a cone. Mapping a cone surface requires a 2D set of k -points around K .

Graphene DOS and band-path tools

- For DOS, perform a dense $N \times N \times 1$ NSCF calculation and use `dos.x`. A small Gaussian width is an accessible starting route.
- Increase N and reduce the broadening together. The apparent DOS at the Dirac point is particularly sensitive to both.
- Path tools such as SeeK-path or XCrySDen help identify high-symmetry points. Verify their coordinate convention and any cell standardization.
- A path generated for a rotated or standardized cell must be converted before use with your original cell.

A smooth-looking curve can hide insufficient sampling. The energy-grid spacing in the postprocessor does not generate additional eigenvalues.

Reference: [Materials Cloud SeeK-path](#)

A 3×3 graphene supercell

Replicate the primitive cell three times along each in-plane direction:

$$\mathbf{A}_1 = 3\mathbf{a}_1, \quad \mathbf{A}_2 = 3\mathbf{a}_2, \quad \mathbf{A}_3 = \mathbf{a}_3.$$

	Primitive cell	3×3 cell
In-plane A	2.46 Å	7.38 Å
Cell height C	20 Å	20 Å
Atoms	2	18
Example equivalent meshes	$18 \times 18 \times 1$	$6 \times 6 \times 1$

Both meshes here are unshifted. The original $9 \times 9 \times 1$ and $3 \times 3 \times 1$ pair is another equivalent pair, subject to convergence testing.

Generating all 18 supercell coordinates

For each primitive fractional position (x, y, z) , create

$$\left(\frac{x+i}{3}, \frac{y+j}{3}, z\right), \quad i, j \in \{0, 1, 2\}.$$

This Python snippet prints a valid position card:

```
basis = [(0.0, 0.0, 0.5), (1/3, 2/3, 0.5)]
print("ATOMIC_POSITIONS crystal")
for i in range(3):
    for j in range(3):
        for x, y, z in basis:
            print(f"C {(x+i)/3:.10f} {(y+j)/3:.10f} {z:.10f}")
```

- Set nat=18, A=7.38, C=20, a new prefix, and the $6 \times 6 \times 1$ mesh.
- For equivalent empty-band coverage, scale nbnd=8 to nbnd=72.
- If you use celldm, the new c/a is one third of its old value, because c stays fixed.

Energy scaling and band folding

For pristine replicas with equivalent meshes and identical numerical settings:

$$E_{\text{SC}} \simeq 9E_{\text{PC}}, \quad \frac{E_{\text{SC}}}{18} \simeq \frac{E_{\text{PC}}}{2}.$$

- The reciprocal cell shrinks by three in each in-plane direction. Primitive-cell states fold into this smaller Brillouin zone.
- A supercell has more bands at each k -point because it contains more atoms and electrons.
- Folded bands do not by themselves imply new physics. A defect or distortion breaks primitive translational symmetry and can change the spectrum.
- Compare energy per atom. Finite grids and stopping thresholds can leave small numerical differences.

Adsorption energies require balanced compositions and consistent reference states.

What a relaxation actually minimizes

For fixed lattice vectors, atomic forces are

$$\mathbf{F}_I = -\frac{\partial E}{\partial \mathbf{R}_I}.$$

1. Converge the electronic density at the current atomic positions.
2. Calculate forces.
3. Move the atoms using the optimizer, then repeat.
 - `calculation='relax'` changes positions at fixed cell.
 - `calculation='vc-relax'` can also change cell vectors, subject to `cell_dofree`.
 - A converged relaxation finds a stationary configuration near the starting structure. It does not prove a global minimum or dynamical stability.

Relaxation controls and stopping criteria

For a fixed-cell relaxation, edit &CONTROL and append &IONS:

```
! Inside &CONTROL:  
calculation='relax'  
etot_conv_thr=1.0d-5  
forc_conv_thr=1.0d-4  
nstep=100  
  
! After &ELECTRONS:  
&IONS  
  ion_dynamics='bfgs'  
/
```

The energy threshold is in
Ry/cell.

The force threshold is in
Ry/bohr:

$$10^{-4} \frac{\text{Ry}}{\text{bohr}} \simeq 0.00257 \frac{\text{eV}}{\text{\AA}}.$$

QE checks force components and the energy change. Reaching nstep is not convergence.

Use electronic thresholds tight enough that SCF noise is well below the target force. Recompute the final geometry with tighter settings to verify it.

Graphane: hydrogen above and below the sheet

The **chair conformer** of graphane has one hydrogen per carbon, with neighboring H atoms on opposite sides.

- The primitive cell contains $2\text{C} + 2\text{H}$: nat=4, ntyp=2.
- Hydrogenation favors a buckled carbon sheet. Start with small opposite carbon displacements rather than forcing a planar geometry.
- Add a compatible H pseudopotential and reconverge the combined system.
- A fixed-cell relaxation finds positions for the chosen in-plane lattice constant. Scan A or use a suitable in-plane cell constraint to find the equilibrium lattice constant.

Keep the vacuum dimension fixed. A neutral symmetric chair structure differs electrostatically from one-sided hydrogenation or oxygen adsorption.

Graphane: a concrete starting geometry

Use the graphene template with $A=2.54$, $C=20$, $nat=4$, $ntyp=2$, $nbnd=10$, `prefix='graphane'`, and the relaxation controls. Replace the species and position cards:

```
ATOMIC_SPECIES
C 12.011 C.UPF
H  1.008 H.UPF
ATOMIC_POSITIONS angstrom
C 0.0 0.0000000000 10.20
C 0.0 1.4664696837  9.80
H 0.0 0.0000000000 11.30
H 0.0 1.4664696837  8.70
```

Here $y_2 = A/\sqrt{3}$, carbon buckling is $0.40 \text{ \AA}$, and each initial C–H distance is $1.10 \text{ \AA}$. These initialize the optimizer and are not final bond lengths. Retain a tested in-plane mesh and reconverge cutoffs and occupations.

An epoxide bridge on graphene

An oxygen atom bridging a C–C bond is one useful model of oxygen functionalization.

- Begin with the 3×3 graphene supercell: $18\text{C} + 1\text{O}$, so $\text{nat}=19$, $\text{ntyp}=2$.
- Place O above the midpoint of a neighboring C–C bond, rather than above an arbitrary pair.
- For $d_{\text{C-C}} = a/\sqrt{3}$ and a trial C–O distance r , the initial height above the flat sheet is

$$h = \sqrt{r^2 - (d_{\text{C-C}}/2)^2}.$$

- With $a = 2.46 \text{ \AA}$, $r = 1.50 \text{ \AA}$, this gives $h \simeq 1.321 \text{ \AA}$.

This model does not establish the first step of an oxidation reaction. Reaction pathways and kinetic barriers require separate calculations.

Epoxide input and finite-size checks

Use the 18-atom fractional card from the generator. Add:

```
! Add to ATOMIC_SPECIES:  
O 15.999 0.UPF  
  
! Append to ATOMIC_POSITIONS crystal:  
O 0.0555555556 0.1111111111 0.5660624704
```

- These coordinates put O above the bond joining the first primitive pair, with $C = 20 \text{ \AA}$.
- Set `nat=19`, `ntyp=2`, a new prefix such as `epoxide`, and `calculation='relax'`. Retest bands and cutoffs after adding O.
- A one-sided adsorbate can create a dipole. Check vacuum convergence and use a documented electrostatic treatment when needed.
- The defect repeats in each supercell. Test larger lateral cells before treating it as an isolated defect.

Inspect C–O distances, carbon buckling, maximum force and whether the intended bridge survived.

Troubleshooting by the symptom

Symptom	First checks
UPF not found	Exact filename, working directory, <code>pseudo_dir</code> , read permission
SCF oscillates	Geometry/overlaps, occupations, empty bands, mixing, cutoffs
Bands cannot read data	Same prefix/outdir, successful SCF, unchanged structure/model
Noisy forces or stress	SCF threshold, both cutoffs, mesh and smearing
Relaxation ends early	Convergence message, remaining force, <code>nstep</code> , optimizer/SCF errors
Unexpected band gap	Path/mesh, band count, energy reference, symmetry and model

Keep the input and full output for diagnosis. Create a fresh case-specific prefix instead of deleting useful restart data blindly.

Five checks you can answer without running QE

1. A two-Si cell has eight valence electrons with $n_{\text{spin}}=1$. How many occupied bands are needed?
2. Two cutoff tests differ by 2×10^{-4} Ry per two-atom cell. What is the difference in meV/atom?
3. A pristine 4×4 graphene supercell uses an equivalent mesh to a primitive $24 \times 24 \times 1$ mesh. What are its atom count and mesh?
4. A calculation has $M_{\text{tot}} = 0$ but $M_{\text{abs}} > 0$. Must it be nonmagnetic?
5. A graphene relax run has small forces. Have you found the equilibrium in-plane lattice constant?

Answers and useful sanity checks

1. **Four occupied bands**, since each nonmagnetic band can hold two electrons. Add empty bands for conduction states.
2. 1.361 meV/atom : $2 \times 10^{-4} \times 13.605693 \times 1000/2$.
3. **32 atoms and a $6 \times 6 \times 1$ mesh**, with compatible offsets. The area and electron count increase by 16.
4. **No.** Opposite spin polarization can cancel in the cell integral, as in an AFM state.
5. **No.** `relax` holds the cell fixed. The in-plane lattice parameter needs its own optimization.

Check units, normalization per cell/per atom, and which degrees of freedom your calculation actually allows.

A practical completion checklist

Before reporting a result, you should be able to show:

- The exact structure, QE version, UPFs, XC and spin/occupation model
- SCF convergence and a systematic test of cutoffs and k -sampling for the reported observable
- Smearing and vacuum/defect-separation tests where relevant
- A stated energy reference for bands/DOS, or balanced reference states for energy differences
- For optimized structures, final coordinates, residual forces and relevant stress/cell constraints

Keep complete inputs and outputs. Reproducibility needs the actual settings and pseudopotentials, not just a screenshot of a plot.

Source and original references

Source tutorial

Adapted from the supplied `handson_pwscf.pdf`, *Quantum ESPRESSO tutorial: Self-Consistent Calculations, Supercells, Structural Optimization*, 32 pages. PDF creation date: 24 March 2018. No author is named in the supplied file.

Original methodological references retained

D. J. Chadi and M. L. Cohen, *Special Points in the Brillouin Zone*, Physical Review B **8**, 5747 (1973). [doi:10.1103/PhysRevB.8.5747](https://doi.org/10.1103/PhysRevB.8.5747)

H. J. Monkhorst and J. D. Pack, *Special points for Brillouin-zone integrations*, Physical Review B **13**, 5188 (1976). [doi:10.1103/PhysRevB.13.5188](https://doi.org/10.1103/PhysRevB.13.5188)

Quantum ESPRESSO overview

P. Giannozzi et al., J. Phys.: Condens. Matter **21**, 395502 (2009). [Open-access paper](#)

Official documentation and further reading

- `pw.x`: syntax, lattice vectors, occupations, cutoffs and relaxation
- `bands.x`: band ordering, symmetry and output files
- `dos.x`: DOS integration and energy units
- `projwfc.x`: projections, PDOS formats and integration
- PWscf guide: structural optimization
- Pseudopotential sources and validation guidance
- SeeK-path: reciprocal-space paths and cell conventions

Online documentation consulted on 2 October 2026, reporting QE 7.5. Use your installed version's manuals for version-specific behavior and cite the pseudopotential datasets used in your own work.